EPFL - Computer Science

COM-407 Advanced Networks

Lecture 14 — December 18, 2025

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External Routing

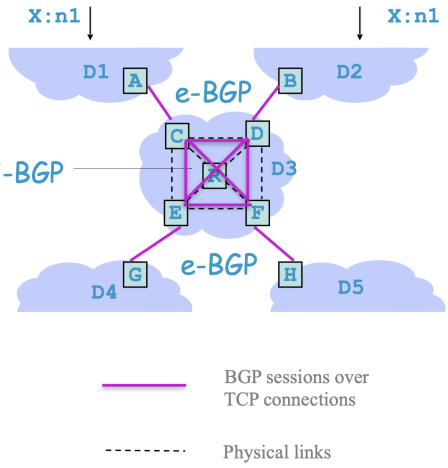
BGP

1. How does BGP work?

- BGP routers/*speakers* talk to each other *over TCP* connections
- Each BGP router [BGP-4, RFC 4271]:
  - receives and stores *candidate routers* from its BGP neighbors, after applying *import* policy rules
  - applies the *decision process* to *select at most one route* per destination prefix and keeps all other imported routers as backup
  - exports the *selected* routes to BGP neighbors, after applying *export* policy rules and possibly *aggregation*
- Routers are advertised via **UPDATE** messages that contain *only modifications* = **new paths, withdrawals**
- Other messages: **OPEN** (=sync after boot-up), **NOTIFICATION** (= reset), **KEEPALIVE** (= periodic beacon to notify BGP peers that BGP on router is running); absence of KEEPALIVE after some time triggers a withdrawal UPDATE

2. Types of BGP (used in parallel)

- e(xternal)-BGP:
  - routing exchange between routers in *different* domains (i.e border routers across neighbor domains)
- i(nternal)-BGP:
  - routing exchange between routers *within* same domain (i.e. domain's border or other internal BGP-speaking routers)



In i-BGP, BGP speakers/peers:

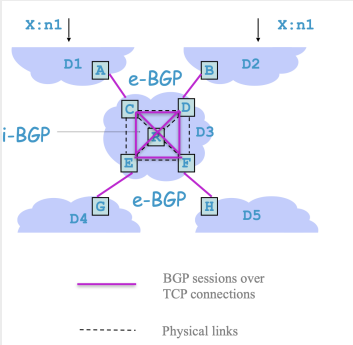
- form a BGP mesh network (all-to-all)
- advertize routes learn from e-BGP but they never:
  - repeat routes learnt from i-BGP to avoid redundant traffic
  - prepend own AS number over i-BGP
  - modify the "NEXT-HOP" attribute of a route
- know about inter-domain link subnets via IGP

Say what is always true
1. Two BGP peers must be connected by a TCP connection.
2. Two BGP peers must be "on link" (on the same subnet)

A. 1
B. 2
C. 1 and 2
D. None
E. I don't know

Answer: A. Internal peers need not be "on link". BGP peers communicate (typically) with TCP.

Which BGP updates may be sent?

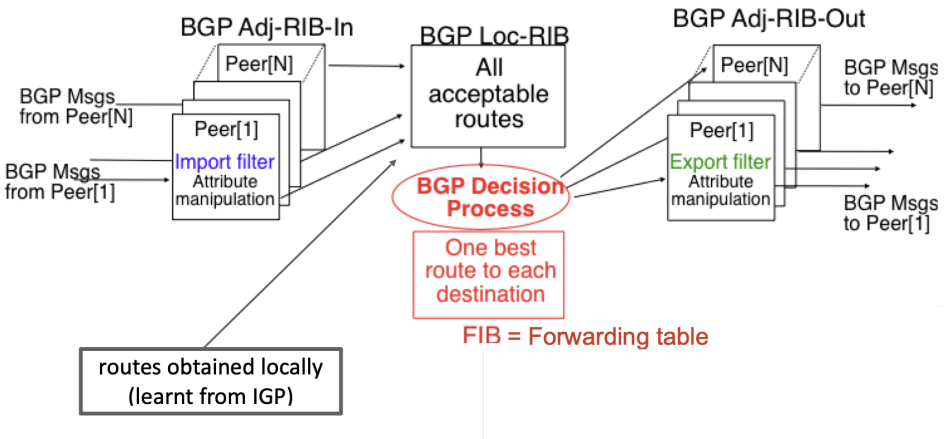


1. C ⇒ A AS-PATH = [D3 D2 X : n1]
2. D ⇒ E AS-PATH [D2 X : n1]
3. C ⇒ E AS-PATH = [D2 X : n1]

A. 1
B. 2
C. 3
D. 1 and 2
E. 1 and 3
F. 2 and 3
G. All
H. None
I. I don't know

Answer: D. The router C ⇒ E was learnt via internal BGP. Therefore, it should *not be re-advertized* over i-BGP.

Operation of a BGP router

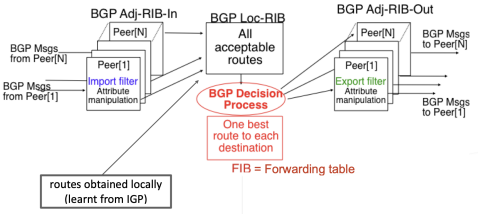


RIB - Route Information base,
IGP - Internal Gateway Protocol

Routes, RIBs, Routing Table

An advertised *route* has several attributes:

- **destination** (subnet) prefix
- **path** to destination
- **NEXT-HOP** (modified by e-BGP, left unchanged by i-BGP
- **ORIGIN**: route learnt from IGP, BGP, static
- Other attributes...



Routers + their attributes are stored in the Routing Information Bases (RIBs):

Ajd-RIB-in, Loc-RIB, Adj-RIB-out

Like any IP host or router, a BGP router also has an IP Forwarding Table used for packet forwarding, in real time

The Decision Process

The decision process chooses *at most one route* to each different destination *prefix* as best

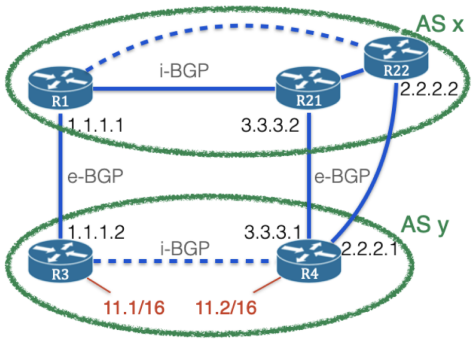
How?

- A route can be selected only if its next-hop is reachable
- For each dest prefix, all acceptable routes are compared w.r.t. their attributes using a sequence of criteria (until only one route remains); a typical sequence is:
  0. Highest weight (cisco proprietary)
  1. Highest LOCAL-PREF
  2. Shortest AS-PATH
  3. Lowest MED (multi-exit discriminator), if taken seriously this domain
  4. e-BGP > i-BGP
  5. Shortest path to NEXT-HOP, according to IGP
  6. Lowest BGP identifier
- The result of the decision process is stored in forwarding talbe and in Adj-RIB-out (at most *one route per destination per BGP peer*)
- The router sends updates when Adj-RIB-out *changes* (addition or withdrawal) after applying *export rules*

Fundamental Example

- 4 BGP routers connected directly (solid lines) or indirectly (dash lines), speaking both e-BGP and i-BGP;
- 2 Ases, **x** and **y**, each one running its own IGP.
- Assume R3 and R4 are configured to advertize both prefixes of **y**.

We focus on R1 and show its BGP information bases



Another Fundamental Example

- 3 BGP routers in AS **x**, also running an OSPF
- Assume:
  - all link costs are equal to 1. R3 and R4 advertize *only* their directly attached prefixes.

We focus on R1 and show it's BGP information bases

How are routers originated (=sourced)?

Several methods for sourcing a route:

- Static configuration: manually tell this BGP router which prefixes to originate ("network"command in FRR)
- Redistribute connected: tell this BGP router to originate *all directly attached prefixes* (all routers in network may run i-BGP, no need for IGP in this case)
- Redistribute from IGP: tell this BGP router to originate *all prefixes learnt by IGP*, e.g.: redistribute OSPF into BGP
  - if IGP=OSPF, in principle, *only internal prefixes* should be redistributed
  - Such BGP routes have attribute ORIGIN=IGP
  - When originated, the BGP NEXT-HOP of such a route is its IGP next-hop.

Redistribution means that we redistribute information learn by BGP to IGP and from IGP to BGP!

2. Aggregation (of routes)

Routes usually *overlap*

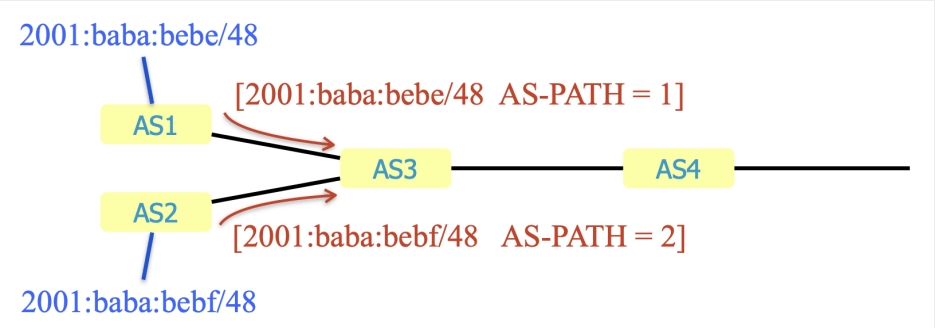
- expected very frequently in IPv6 (recall the way we delegate prefixes),
- less frequently in IPv4

So, *prefix aggregation* can reduce the number of routes

- In IP forwarding tables
- In BGP advertisements

Otherwise several hundreds of thousands of entries or advertizements (e.g., consider transit ISPs that rarely have a default route to 0.0.0.0)

Can AS3 aggregate these routes into a single one?



A. Yes and the aggregated prefix is 2001:baba:bebe/47
B. Yes and the aggregated prefix is 2001:baba:bebf/48
C. Yes but the aggregated prefix is none of the above
D. No
E. I don't know
Answer: A. The two prefixes are contiguous and ca be aggregatedas 2001:baba:bebe/47

3. How routes learnt by BGP are written into Forwarding Tables?

Two ways:

1. *Redistribution of BGP into IGP*: route learnt by BGP is passed to IGP (e.g., OSPF)
   - Typically only routers learnt by e-BGP are redistributed (unless BGP redistribute-internal is used)
   - IGP propagates routing information to all routers within domain
   - Works with OSPF, might not work with other IGPs (tables too large for IGP)

2. Injection **Destination Prefix and NEXT-HOP of BGP route is directly copied into forwarding table**
   - Why used? IGP avoids dealing with a large number of table entries (consider convergence issues in distance-vector algorithms, such as RIP).
   - Injection helps only the particular BGP router; Routing information is not propagated to other intra-domain routers.
   - Typically used in Cisco routers and FRR.

Example (re-distribution)

Example (injection)

Final Exam

The last exam is going to be related to the last years.

Upload 2 different versions: a blank and one with solutions. Start with the blank one is **super important!**

Need to practice to finish the exam on time. The course is "easy", but the examen can only be finished with practice.

There will be a cheatsheet 3 A4 sheets (6 pages). Slides authorized, everything!

QA session in the last week. Tuesday is a good time.